Universidad Tecnológica de Jalisco
Innovación y Excelencia

TÉCNICO SUPERIOR UNIVERSITARIO EN TECNOLOGÍAS DE LA INFORMACIÓN ÁREA DESARROLLO DE SOFTWARE MULTIPLATAFORMA

NOSQL

Quinto A

ENTREGABLE
PYTHON VS MONGO ATLAS

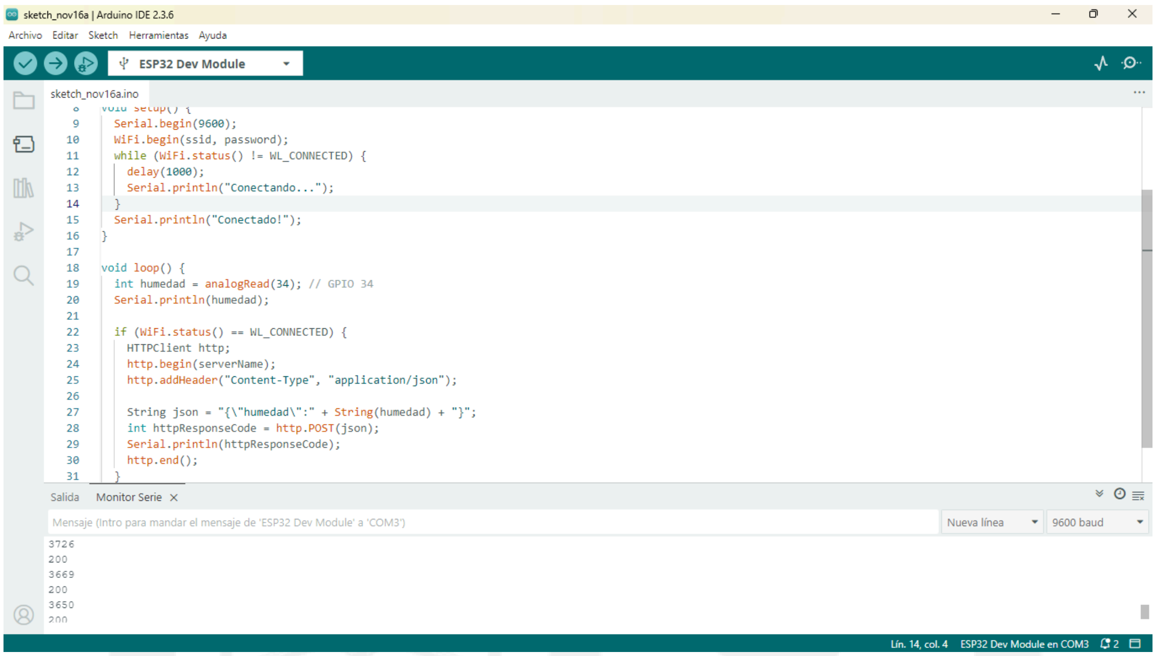
PRESENTA:

Luis Rodolfo Silva Castellanos

PROFESOR:
CARLOS IVAN

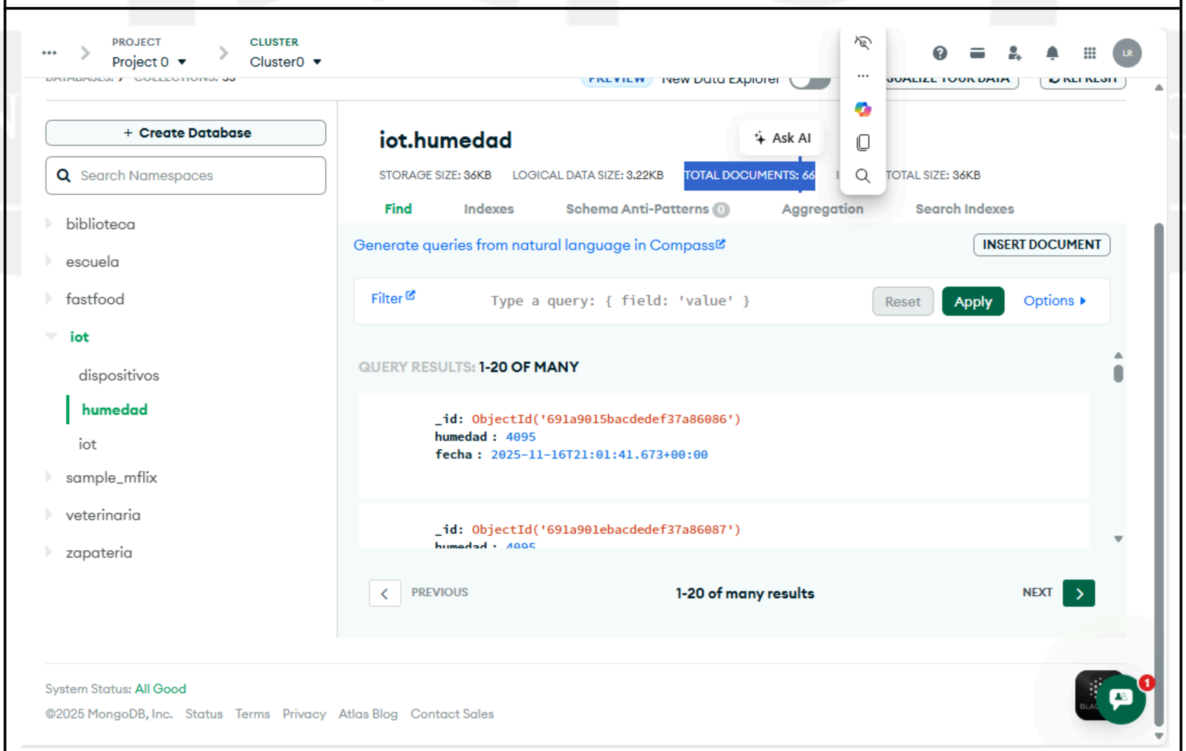
GUADALAJARA, JALISCO, 16 DE NOVIEMBRE DE 2025

CÓDIGO ARDUINO




```
#include <WiFi.h>
#include <HTTPClient.h>

const char* ssid = "Laimposible2.4";
const char* password = "H2Oco2HJems";
const char* serverName = "http://192.168.1.87:5000/humedad";

void setup() {
  Serial.begin(9600);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Conectando...");
  }
  Serial.println("Conectado!");
}
```

```
void loop() {
  int humedad = analogRead(34); // GPIO 34
  Serial.println(humedad);

  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;
    http.begin(serverName);
    http.addHeader("Content-Type", "application/json");

    String json = "{\"humedad\":" + String(humedad) + "}";
    int httpResponseCode = http.POST(json);
    Serial.println(httpResponseCode);
    http.end();
  }

  delay(5000);
}
```



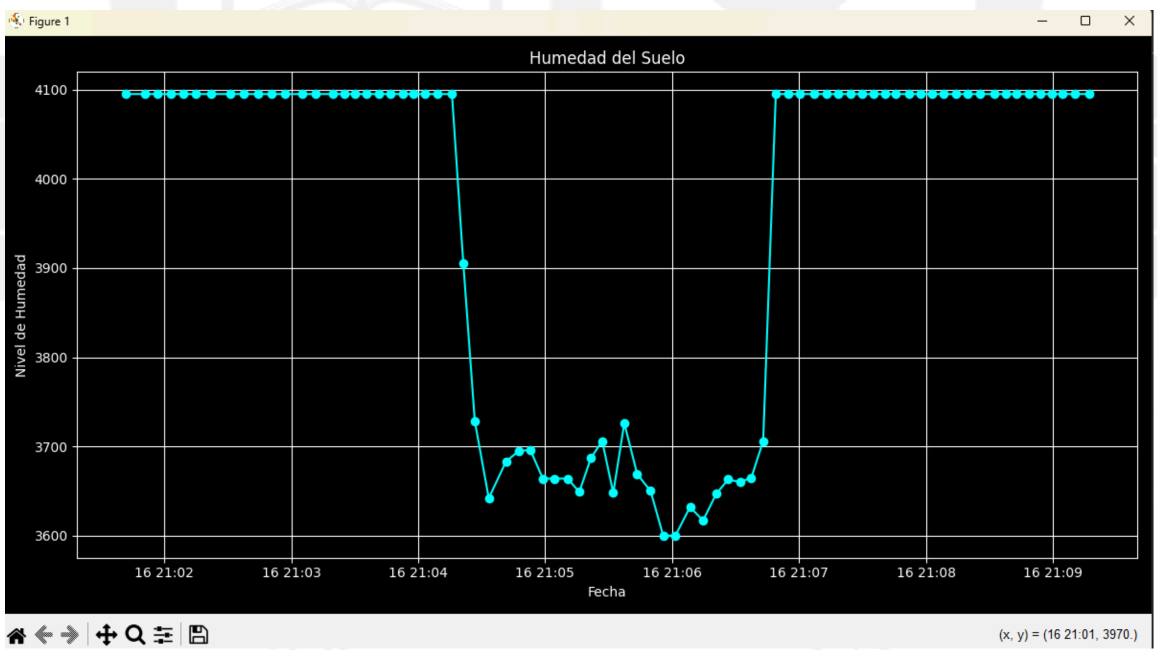
CODIGO PARA HACER EL POST A MONGO BD ATLAS

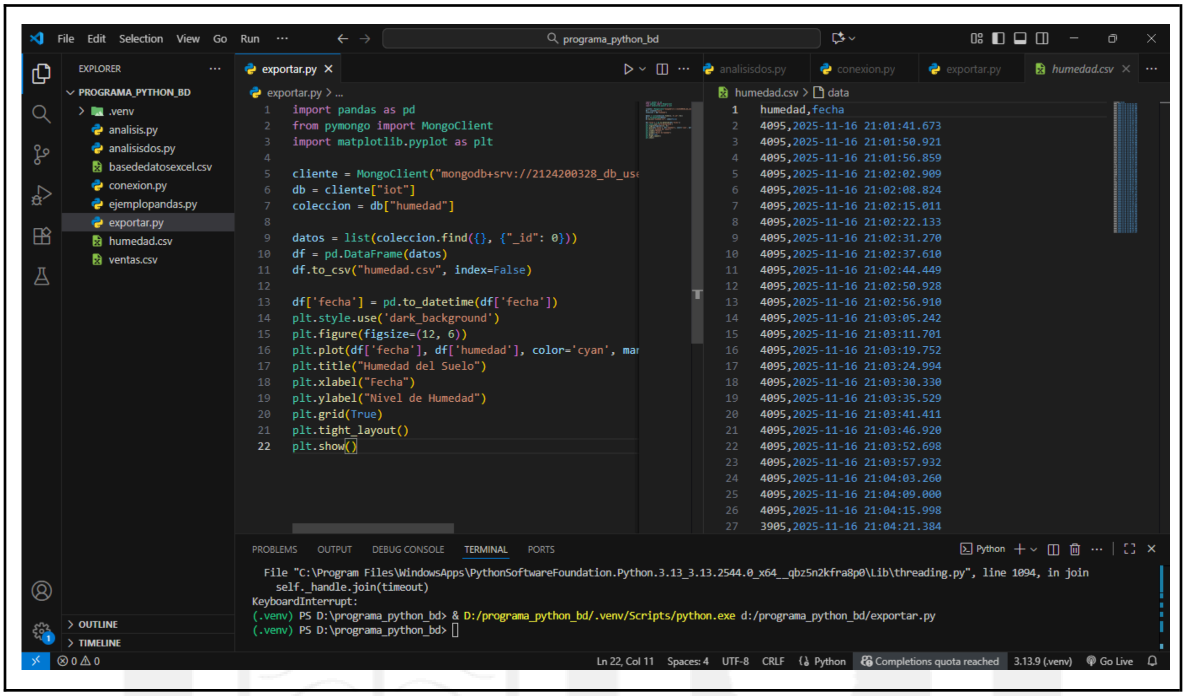
```python
from flask import Flask, request
from pymongo import MongoClient
from datetime import datetime

app = Flask(__name__)
cliente =
MongoClient("mongodb+srv://2124200328_db_user:goTe8610@cluster0.hr2u
uz8.mongodb.net/?appName=Cluster0")
db = cliente["iot"]
coleccion = db["humedad"]

@app.route('/humedad', methods=['POST'])
def recibir():
    data = request.get_json()
    data["fecha"] = datetime.now()
    coleccion.insert_one(data)
    return {"status": "ok"}

app.run(host='192.168.1.87', port=5000)
```


CÓDIGO PARA IMPORTAR CSV Y GRAFICAR

```python
import pandas as pd
from pymongo import MongoClient
import matplotlib.pyplot as plt

cliente =
MongoClient("mongodb+srv://2124200328_db_user:goTe8610@cluster0.hr2u
uz8.mongodb.net/?appName=Cluster0")
db = cliente["iot"]
```

```python
coleccion = db["humedad"]

datos = list(coleccion.find({}, {"_id": 0}))
df = pd.DataFrame(datos)
df.to_csv("humedad.csv", index=False)

df['fecha'] = pd.to_datetime(df['fecha'])
plt.style.use('dark_background')
plt.figure(figsize=(12, 6))
plt.plot(df['fecha'], df['humedad'], color='cyan', marker='o')
plt.title("Humedad del Suelo")
plt.xlabel("Fecha")
plt.ylabel("Nivel de Humedad")
plt.grid(True)
plt.tight_layout()
plt.show()
```




programa_python_bd
EXPLORER
PROGRAMA_PYTHON_BD
.venv
analisis.py
analisisdos.py
basededatosexcel.csv
conexion.py
ejemplopandas.py
exportar.py
humedad.csv
ventas.csv
exportar.py
import pandas as pd
from pymongo import MongoClient
import matplotlib.pyplot as plt
cliente = MongoClient("mongodb+srv://2124200328_db_us
db = cliente["iot"]
coleccion = db["humedad"]
datos = list(coleccion.find({}, {"_id": 0}))
df = pd.DataFrame(datos)
df.to_csv("humedad.csv", index=False)
df['fecha'] = pd.to_datetime(df['fecha'])
plt.style.use('dark_background')
plt.figure(figsize=(12, 6))
plt.plot(df['fecha'], df['humedad'], color='cyan', ma
plt.title("Humedad del Suelo")
plt.xlabel("Fecha")
plt.ylabel("Nivel de Humedad")
plt.grid(True)
plt.tight_layout()
plt.show()
humedad.csv
humedad,fecha
4095,2025-11-16 21:01:41.673
4095,2025-11-16 21:01:50.921
4095,2025-11-16 21:01:56.859
4095,2025-11-16 21:02:02.909
4095,2025-11-16 21:02:08.824
4095,2025-11-16 21:02:15.011
4095,2025-11-16 21:02:22.133
4095,2025-11-16 21:02:31.270
4095,2025-11-16 21:02:37.610
4095,2025-11-16 21:02:44.449
4095,2025-11-16 21:02:50.928
4095,2025-11-16 21:02:56.910
4095,2025-11-16 21:03:05.242
4095,2025-11-16 21:03:11.701
4095,2025-11-16 21:03:19.752
4095,2025-11-16 21:03:24.994
4095,2025-11-16 21:03:30.330
4095,2025-11-16 21:03:35.529
4095,2025-11-16 21:03:41.411
4095,2025-11-16 21:03:46.920
4095,2025-11-16 21:03:52.698
4095,2025-11-16 21:03:57.932
4095,2025-11-16 21:04:03.260
4095,2025-11-16 21:04:09.000
4095,2025-11-16 21:04:15.998
3905,2025-11-16 21:04:21.384
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
File "C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.13_3.13.2544.0_x64__qbz5n2kfra8p0\Lib\threading.py", line 1094, in join
self._handle.join(timeout)
KeyboardInterrupt:
(.venv) PS D:\programa_python_bd> & D:/programa_python_bd/.venv/Scripts/python.exe d:/programa_python_bd/exportar.py
(.venv) PS D:\programa_python_bd>
Ln 22, Col 11 Spaces: 4 UTF-8 CRLF Python Completions quota reached 3.13.9 (.venv) Go Live